SEMIN JIN

AI DESIGN COMPUTING & HUMAN-CENTERED DESIGN EVALUATION

SEMINJIN.COM

Postdoctoral Researcher · Design AI Lab
Human-Centered AI Design Institute, Hanyang University
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Design computing researcher advancing AI-based generation and human-centered design evaluation. Grounded in interior and spatial design, my work investigates how bodies use 3D environments, how combinations of color, material, and lighting shape impressions and preferences, and how generative AI influences designers' exploration and decisions. I combine LLMs, diffusion and vision-language models with anthropometric simulation, motion capture, interactive systems, and user studies. Across these areas, I translate insights from design practice into computational methods for human-usable spatial generation, meaningful visual exploration, and empirical analysis of AI-mediated creative processes.

EDUCATION

Sep 2024 – Aug 2026

Doctor of Philosophy in Interior Architecture Design
Hanyang University, Seoul · Graduate School of Interior Architecture Design
Advisor: Prof. Kyung Hoon Hyun | **GPA 4.41 / 4.50 · 38 credits**
Doctoral Dissertation: *Anthropometric Simulation and Human Pose Evaluation for Generative Spatial Design.*

Mar 2022 – Aug 2024

Master of Science in Interior Architecture Design
Hanyang University, Seoul · Graduate School of Interior Architecture Design
Advisor: Prof. Kyung Hoon Hyun | **GPA 4.44 / 4.50 · 27 credits**
Master's Thesis: *Data-Driven Aesthetic Assessment Systems for Supporting Interior Design.*

Mar 2018 – Feb 2022

Bachelor of Arts in Architectural Interior Design
Kaywon University of Art & Design, Uiwang
Top-up program (2021 – 2022) | **GPA 4.23 / 4.50 · 23 credits**
Preceded by Associate of Arts (A.A.) in Interior Architecture Design, 3-yr (2018 – 2021) | **GPA 4.09 / 4.50 · 122 credits**

PUBLICATIONS

Impact Factors based on 2024 JCR · H5-index from Google Scholar

UNDER REVIEW

[U1] **Jin, S.**, ..., & Hyun, K. H. Biomechanical Simulation for Evaluating and Authoring Human-Usable Spatial Designs. Submitted to ***Automation in Construction***.
SCIE

[U2] ..., ..., ..., **Jin, S.**, ..., ..., & Hyun, K. H. Design Workflow-Aware Agentic AI for Designer-Centered Design Generation. Submitted to ***International Journal of Human–Computer Interaction***.
SCIE

JOURNAL ARTICLES

[1] **Jin, S.**, & Hyun, K. H. (2026). Human-operational 3D indoor layout generation with LLM-driven anthropometric simulation. ***Automation in Construction***, 184, 106846.
SCIE | IF 11.5 | TOP 1% | First author

[2] **Jin, S.**, & Hyun, K. H. (2026). Blending designer insights with predicted crowd-based color compatibility in interior color design. ***Color Research and Application***.

SCIE | IF 1.4 | First author

[3] Seo, J., **Jin, S.**, Choi, K., Hyun, K. H., & Joung, J. (2026). An explainable AI-based approach for identifying interior design style principles. ***Advanced Engineering Informatics***, 69, 103879.

SCIE | IF 9.9 | TOP 2%

[4] **Jin, S.**†, Choi, J.†, & Hyun, K. H. (2025). CMLsearch: Semantic visual search and simulation through segmented colour, material, and lighting in interior image. ***Journal of Computational Design and Engineering***, 12(1), 179–299.

SCIE | IF 6.1 | TOP 7% | † Co-first author

[5] **Jin, S.**, & Hyun, K. H. (2024). Enhancing the interior design process with crowdsourced furnishing pairing recommendations. ***Archives of Design Research***, 37(2), Vol. 150.

SCOPUS | KSDS BEST PAPER 2024 | First author

CONFERENCE PROCEEDINGS

[6] **Jin, S.**†, Kim, D.†, Ryu, J., & Hyun, K. H. (2026, April). Behavior-Aware Anthropometric Scene Generation for Human-Usable 3D Layouts. In ***Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)***, Barcelona, Spain.

BK TOP-TIER CONF. | H5-INDEX 139 | † Co-first author

[7] Lee, S. W., Jo, T. H., **Jin, S.**, Choi, J., Yun, K., Bromberg, S., Ban, S., & Hyun, K. H. (2024). The impact of sketch-guided vs. prompt-guided 3D generative AIs on the design exploration process. In ***Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '24)***, Honolulu, HI, USA.

BK TOP-TIER CONF. | H5-INDEX 139

PATENTS

REGISTERED	**인테리어 제품 검색 서비스 제공 방법 및 장치** Patent No. KR 10-2943193, Registered March 19, 2026 · 현경훈, **진세민**, 최지인
FILED	**인체측정 데이터 기반 3차원 실내 공간 레이아웃 생성 및 사용성 검증 시스템 및 방법** Application No. KR 10-2026-0040371 · 현경훈, **진세민**, 김동혁
FILED	**다중 에이전트를 기반으로 창의적 아이디어를 구조화하고 시각화하는 시스템 및 방법** Application No. KR 10-2026-0036677 · 현경훈, **진세민**, 이승원
FILED	**디자인 아티팩트 기반 이미지 생성 시스템 및 방법** Application No. KR 10-2025-0184966 · 현경훈, 이승원, 최지인, **진세민**
FILED	**군중 선호도 데이터를 활용한 공간 디자인 이미지의 설명적 평가 및 색채 조합 생성 시스템과 방법** Application No. KR 10-2024-0054887 · 현경훈, **진세민**
DESIGN · REGISTERED	**화면디자인이 표시된 디스플레이 패널 (다중 에이전트 기반 창의적 디자인 대화 구조화 및 시각화 시스템)** Design Registration No. KR 30-1348144, Registered April 23, 2026 · 현경훈, **진세민**, 이승원

EXPERIENCE

RESEARCH

2026 – Present

Postdoctoral Researcher

Design AI Lab · Human-Centered AI Design Institute, Hanyang University

Lead research on generative spatial design — anthropometric and biomechanical human simulation, 3D scene generation, and agentic AI for designer-centered workflows.

2022 – 2026

Graduate Researcher

Design AI Lab · Human-Centered AI Design Institute, Hanyang University

Conducted research on human simulation–driven 3D scene generation and AI-assisted design systems.

Led proposal development and research execution for a national project on agentic AI that integrates generation and evaluation of human-aware 3D spatial layouts (National Research Foundation of Korea · Core Research Program).

Contributed to a large-scale industry–academia consortium building designer-centered AI generation pipelines grounded in real design workflows (Korea Evaluation Institute of Industrial Technology · Design Tech Program).

TEACHING

Fall 2026

Instructor

Graduate School of Interior Architecture Design, Hanyang University

Course: *Design Computing Research* (Graduate, English Course)

Spring 2025, Spring 2026

Teaching Fellow — 4-week module

Department of Interior Architecture Design, Hanyang University

Course: *Furniture Design and Digital Fabrication* (English Course)

Led hands-on tutorials in 3D printing, laser cutting, and CNC fabrication, and provided project-based design critiques.

EXHIBITIONS

Apr 2026

Designing Layouts from Human Motion

38th International Invitational Exhibition of Spatial Design · Korean Institute of Spatial Design (KISD), Seoul · Kim, D., **Jin, S.**, & Hyun, K. H.

Nov 2023

Sketch to 3D Generation

KIID International Invitation Exhibition, Seoul · **Jin, S.**, Jo, T. H., & Hyun, K. H.

Nov 2023

Dynamic Office Design for Brand Innovation

KIID International Invitation Exhibition, Seoul · **Jin, S.**, & Hyun, K. H.

Nov 2022

Return to Flow of Nature

30th Anniversary International Invitational Exhibition · Korean Institute of Interior Design (KIID), Seoul · **Jin, S.**, & Hyun, K. H.

AWARDS, CERTIFICATIONS & SERVICE

AWARDS

Aug 2026

Outstanding Doctoral Dissertation Award

Hanyang University · Graduate School of Interior Architecture Design

For *Anthropometric Simulation and Human Pose Evaluation for Generative Spatial Design.*

Nov 2024

Best Paper Award · 우수논문상

Korean Society of Design Science (KSDS) · 2024 Annual Conference

For “Enhancing the Interior Design Process with Crowdsourced Furnishing Pairing Recommendations,” *Archives of Design Research*, Vol. 37, No. 2.

Nov 2022

Excellence Award · 우수작품상

Korean Institute of Interior Design (KIID) · 30th Anniversary International Invitational Exhibition

Work: *Return to Flow of Nature.*

EARLIER

2020

Finalist · 입선 — 2nd Universal Design Idea Competition
· 제2회 유니버설디자인 아이디어 공모전

2019

Encouragement Award · 장려상 — Space Design Grand Prix · 공간디자인대전

CERTIFICATIONS

2021

Engineer Colorist · 컬러리스트기사

Korea National Technical Qualification · Human Resources Development Service of Korea

ACADEMIC SERVICE

Apr 2026

Position Paper — Beyond Semantic Similarity: Open Challenges for Embedding-Based Creative Process Analysis Across AI Design Tools

Herding Cats Workshop · ACM CHI 2026, Barcelona, Spain

Oct 2025

Demo Presentation

Mind & Motion Workshop · ACM UIST 2025, Busan, Korea

2024 – Present

Reviewer

Conferences: ACM CHI · ACM UIST · ACM Creativity & Cognition (C&C)

Journals: *Frontiers of Architectural Research* (2026) · *Archives of Design Research*